



UNIVERSITY
of HULL

Computer Science

BSc / MEng Artificial Intelligence

October 2026

Artificial Intelligence

Shaded modules are offered as options on your programme, all other modules are compulsory.

Level 4 (Certificate Stage)

Trimester 1	Trimester 2
Programming Portfolio (40cr. core)	
Algorithms and Data Structures	Databases and the Cloud
Professional Development	Computational Thinking

Common first year allows for easy transfer between the available programmes.

Level 5 (Diploma Stage)

Trimester 1	Trimester 2
Design, Develop, Deploy (40cr. core)	
Artificial Intelligence	Data Analysis and Visualisation
Foundations of Cybersecurity and Secure Development	Distributed Systems Programming

Post-Diploma Stage (optional)

Trimester 1	Trimester 2
Year in Industry/Year Abroad	

Level 6 (Honours Stage)

Trimester 1	Trimester 2
Honours Stage Project (40 credits)	
Communicating and Teaching Computing (40 credits)	
Computer Vision	Natural Language Processing
Machine Learning	Generative and Collaborative AI

Level 7 (Masters Stage) (optional)

Trimester 1	Trimester 2
Commercial Development Practice (40 credits)	
Component Based Software Engineering Practice	Cutting Edge Concepts
Edge AI	Responsible, Safe and Explainable AI

Please note that the content in the following pages is subject to change, as our curriculum evolves to match the rapidly changing Computer Science landscape.

Level 4

Programming Portfolio

Core tools: C#

Whether you've never written a line of code or you've been tinkering for years, this module meets you where you are. You'll learn to think like a programmer — breaking problems down, writing clean logic, and building working software in C#. By the end of the year you'll have moved from your very first program to designing with classes, objects and inheritance.

You'll explore:

- Core programming building blocks — variables, loops, functions and data types
- Searching, sorting and working with arrays, stacks and queues
- Object-oriented programming — classes, inheritance and encapsulation
- Reading, writing and managing code like a professional, using source control

Algorithms and Data Structures

Every great piece of software starts with a great idea for how to solve the problem — that's what this module teaches you to find. You'll step away from the screen and learn to think in code before you write it, exploring the algorithms and data structures that separate software that merely works from software that works brilliantly.

You'll explore:

- Pseudocode and state machines for planning before you program
- Core data structures, and how to choose the right one for the job
- Sorting and searching algorithms, and how to measure their efficiency
- Design thinking that carries into every other module on your degree

Databases and the Cloud

Core tools: C#, Microsoft SQL Server

Modern software rarely stores its data on just one machine anymore — this module shows you how databases and the cloud work together to make that possible. You'll learn how cloud computing and virtualisation actually work, then design and build real databases, from relational systems through to NoSQL, before bringing the two together to build a working application backed by a database hosted in the cloud.

You'll explore:

- Designing and implementing relational and NoSQL databases
- Virtualisation, containers, and the cloud service models behind them — IaaS, PaaS and SaaS
- Connecting a real application to a cloud-hosted database
- Cloud security, reliability and disaster recover

Computational Thinking

Core tools: programming platforms and mathematical software

Computing is bigger than any one programming language — this module is about how to think as a computer scientist. Drawing on a genuinely global history of innovation, you'll build the mathematical and logical foundations that everything else on your degree rests on.

You'll explore:

- Abstraction and computational methods
- The maths behind computing — algebra, number systems and statistics
- Logic, Boolean algebra and set theory
- Vectors and matrices, and their uses in graphics, AI and data
- Practical aspects of how computers handle different number types and the way that mathematical libraries and tools can help to do the actual calculations

Professional Development

A computer science degree is about more than code — this module helps you build the career to go with it. You'll explore the ethical, legal, social and sustainability challenges facing the sector, and the professional skills that set graduates apart. A hands-on team project puts those skills into practice straight away, while you start building a portfolio that showcases your work.

You'll explore:

- Sector challenges — ethics, diversity, sustainability and cybersecurity
- Communication, research and self-presentation skills
- Real teamwork on a programme-specific group project
- Building your own professional portfolio

Level 5

Design, Develop, Deploy

Core tools: *GitHub*, *GitHub Actions*

This is where you learn to build software the way the industry actually builds it — from first idea to live deployment. As the backbone module of your second year, you'll work through the entire software development lifecycle, solo and as part of a team, picking up the methodologies and tools employers expect you to know.

You'll explore:

- Agile, iterative and waterfall development approaches
- Gathering and modelling real-world requirements
- Test-driven and behaviour-driven development
- Version control, CI/CD and cloud deployment with GitHub Actions

Artificial Intelligence

Learn to build the kind of intelligent systems that are reshaping every industry and understand the ethics behind them. You'll get a practical grounding in how AI actually works, from the history of the field to the algorithms powering it today, building your own solutions to real problem-solving challenges along the way.

You'll explore:

- The history and ethics and trustworthy of artificial intelligence
- Pathfinding and problem-solving search algorithms
- Supervised and unsupervised learning, and neural networks
- Reasoning under uncertainty with Bayesian methods, and an introduction to reinforcement learning
- A brief introduction of fairness, explainability and AI Safety

Data Analysis and Visualisation

Data is only powerful once someone can make sense of it — this module teaches you to be that person. You'll work with real-world data from cleaning and wrangling it through to building and evaluating models, finishing with the skills to turn numbers into a story that's clear, compelling and honest.

You'll explore:

- Data cleaning, exploration and dimensionality reduction with pandas, scikit-learn and NumPy
- Classification, regression and clustering algorithms
- Rigorous experimentation, evaluation and hyperparameter tuning
- Spotting and mitigating bias in data and models
- Visualising data for maximum impact

Distributed Systems Programming

Core tools: .NET Core

Modern software rarely runs on just one machine — learn to build systems that work seamlessly across many. You'll design and build real distributed applications, navigating the unique challenges of concurrency, scalability, performance, security and scale that come with connecting devices and sharing workloads across a network.

You'll explore:

- Architectures and strategies for distributed computing
- Concurrency, security, performance and reliability
- Key distributed design patterns and best practice
- Deploying layered, scalable distributed applications

Foundations of Cybersecurity and Secure Development

Cybersecurity is at the heart of today's digital world. This module provides a broad introduction to the principles, technologies, and practices that protect systems, networks, software, and data from cyber threats. Students will explore how security is integrated into the design and development of digital solutions and gain an understanding of the technical, organisational, and ethical aspects of cybersecurity. The module establishes the essential knowledge required for more advanced study in cybersecurity throughout the degree.

You'll explore:

- Fundamental cybersecurity concepts and security principles
- Cyber threats, vulnerabilities, and risk management approaches
- Secure system and software development practices
- Data protection fundamentals
- Governance, ethics, and legal aspects of cybersecurity
- Emerging trends and technologies in cybersecurity

Year in Industry/Year Abroad

Take everything you've learned so far and put it to work — in a real workplace or a different country altogether. Spend a year gaining genuine professional experience on an industry placement, or broaden your horizons studying at an overseas institution. Either route builds transferable, employability-boosting skills in a credit-bearing year that counts towards your degree.

You'll explore:

- A year of real workplace experience, tailored to your placement
- Or a year studying abroad, guided by department staff to match your goals

Level 6

Honours Stage Project

This is your chance to lead a project entirely your own — from first idea to final report. You'll choose your own topic and see it through from planning to delivery, supported by videos, live sessions and workshops covering everything you need to succeed.

You'll explore:

- Project planning and literature review
- Experimental design, testing and evaluation
- Research ethics and data protection
- Professional report writing and presentation

Communicating and Teaching Computing

Think you understand computer science? Try explaining it to someone else — that's the real test. Spend real time in schools or colleges as a computing ambassador, supported by workshops on teaching practice and the National Curriculum, and discover how explaining your subject deepens your own understanding of it.

You'll explore:

- Classroom management and lesson planning
- How computer science is taught in schools today
- Reflective, experience-based learning from your own placement
- Effective communication skills – a critical life skill whatever career you choose

Generative and Collaborative AI

AI isn't going to replace you — but someone who knows how to work with it might. This module makes sure that's you. You'll build three essential “fusion skills” for working alongside generative AI — asking it the right questions, judging its outputs critically, and teaching it to improve — then apply them to a real project of your own design.

You'll explore:

- Prompting techniques, from chain-of-thought to few-shot learning
- Critically evaluating AI-generated outputs for bias and accuracy
- Ethical, responsible use of generative AI
- A final project building a real generative AI solution

Computer Vision

Teach a computer to see — and to actually understand what it's looking at. From basic image processing through to recognising objects and interpreting entire scenes, you'll build the skills behind everything from facial recognition to self-driving cars.

You'll explore:

- Image processing and feature extraction
- Segmentation, shape analysis and motion tracking
- Object and activity recognition
- The real-world applications and ethics of computer vision

Machine Learning

Move beyond using machine learning models to really understanding how they work. Building on your foundational AI knowledge, you'll dig into the algorithms behind modern machine learning, from classic techniques through to the convolutional networks and generative AI shaping the field today.

You'll explore:

- Gradient descent, regression and regularisation

- Decision trees, random forests and clustering
- Convolutional neural networks
- An introduction to generative AI

Natural Language Processing

Teach a machine to read, understand and write human language — then push it further than you thought possible. You'll explore the techniques behind everything from chatbots to translation tools, working with the large language models and transformer architectures driving the field right now.

You'll explore:

- Large language models and transformer networks
- Generative and conversational AI
- Information retrieval and question answering
- Bias and fairness in language models

Level 7

Commercial Development Practice

Build real software for a real client with industry-standard project management techniques and commercial tools. In Commercial Development Practice, the School's own commercial software development unit, you'll work on genuine client projects that build the business and technical skills employers tell us are actively looking for.

You'll explore:

- Requirements analysis, defining specifications and system design
- Team-based implementation and quality control
- Client liaison, negotiation and project management
- Professional, reflective practice

Component Based Software Engineering Practice

From your phone to the server room, modern software runs on portable and reusable components — learn expert-level techniques to build the platforms that make that possible. Using .NET and the .NET CLR as your guide, you'll explore component-based programming environments and virtual machines, and are design, build and secure your own component-based system.

You'll explore:

- Versioning, metadata and reflection
- Virtual machine design and architecture
- Security in virtual machine execution
- Building a virtual machine

Cutting Edge Concepts

The textbooks can't keep up with computer science — so this module looks at what's happening right now. You'll work with staff to explore genuinely current research across AI, games, security and beyond, learning to critically evaluate academic sources and separate rigorous research from the rest.

You'll explore:

- Identifying high-quality academic sources
- Critically reviewing current research
- Exploring topics beyond the core curriculum
- Skills that carry straight into postgraduate study or research careers

Edge AI

AI doesn't just live in the cloud — learn to run it on the device, in real time, wherever it's needed. You'll design intelligent systems that work under real-world constraints of power, memory and connectivity, building and deploying AI models directly onto edge hardware and connecting them through IoT.

You'll explore:

- Real-time data processing on edge-capable hardware
- Lightweight AI models for devices like Raspberry Pi and Jetson
- IoT integration and sensor networks
- A full end-to-end edge AI project

Responsible, Safe and Explainable AI

The most powerful AI in the world is worthless if nobody can trust it — this module teaches you to build AI people can rely on. You'll explore the cutting edge of responsible, explainable AI, examining how fairness, safety, explainability and transparency can be built into systems from the ground up, not bolted on afterwards.

You'll explore:

- Explainable AI (XAI) and interpretable machine learning
- Detecting and mitigating bias in AI systems
- AI safety, robustness and adversarial attacks
- Responsible multi-agent and autonomous AI systems
- Federated and privacy-preserving learning
- Building AI that is fair, transparent, accountable and trustworthy